CHAROTARUNIVERSITY OF SCIENCE & TECHNOLOGY
1st Semester of B. Pharm. Examination
University Theory Examination April 2016
PH114 Pharmaceutical Chemistry- I

Date: 18.04.2016 , Monday **Time:10.00 a.m. to 1.00 p.m.** **Maximum Marks: 80**

Instructions:

1. **There are three sections in this question paper.**
2. **SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.**
3. **Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.**
4. **Figures to right indicate marks.**
5. **Draw neat sketches wherever necessary.**

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. The 1st law of thermodynamics is
[A] The total energy of an isolated system remains constant though it may change from one form to another
[B] total energy of a system and surrounding remains constant
[C] whenever energy of one type disappears, equivalent amount of another type is produced
[D] all of the above
2. The system in which no thermal energy passes into or out of the system is called
[A] adiabatic system
[B] open system
[C] reversible system
[D] closed system
3. Which form of radioactivity is most penetrating?
[A] alpha particles
[B] beta particles
[C] gama rays
[D] neutrons
4. Radiation is used in the cancer treatment to
[A] destroy cancercausing substances
[B] relieve pain
[C] obtain images of the diseased region
[D] destroy cancer cells
5. "Equal volume of all gases at the same temperature and pressure contain equle number of molecules" is the statement of
[A] combined gas law
[B] Charle's law
[C] Avogadro's law
[D] Boyle's law
6. At constant temperature, the pressure of the gas is reduced to one third, the volume
[A] reduces to one third
[B] increases by three times
[C] remains the same

[D] cannot be predicted

7. The boiling point of the liquid is the temperature at which,
   [A] the vapour pressure of the liquid is equle to the atmosphearic pressure
   [B] the vapour pressure of the liquid is less than the atmosphearic pressure
   [C] the vapour pressure of the liquid is greater than the atmosphearic pressure
   [D] the vapour pressure of the liquid is equle to the square root of atmosphearic pressure
8. Photochemical decomposition of the substance is called
   [A] thermolysis
   [B] thermal dissociation
   [C] photolysis
   [D] hydrolysis
9. The substance which initiates a aphotochemical reaction but itself does not undergo any chemical change is called
   [A] catalysis
   [B] fluorescent
   [C] sensitizer
   [D] none of the above
10. When a non-volatile solute is dissolved in a purew solvent, the vapour pressure of pure solvent
   [A] increases
   [B] decreases
   [C] remains same
   [D] none of these
11. Multi-molecular layers are formed in
   [A] absorption
   [B] physical adsorption
   [C] chemisorption
   [D] reversible adsorption
12. Impurities in pharmaceutical preparation may be due to following sources:
   [A] raw materials
   [B] manufacturing process
   [C] chemical instability
   [D] all of the above
13. Hydrogen peroxide is used as
   [A] antiseptic
   [B] antioxidant
   [C] protective
   [D] acidifying agent
14. Fluoride inhibits caries formation via
   [A] reducing acid solubility of enamel
   [B] bacterial inhibition
   [C] both the above
   [D] increasing acid solubility of enamel
15. Calcium contaning antacid differ from aluminium containing antacid
   [A] depend upon their basic property
   [B] do not have any amphoteric effect
   [C] do not cause systemic alkalosis
   [D] all of the above
16. The category of calcium gluconate is
   [A] antacid
   [B] calcium replenished
   [C] antioxidant
   [D] radiopharmaceuticals
17. Pumice is obtained from__________.
   [A] soil
   [B] volcanic origin
   [C] water

[D] petroleum products

18. Which is the main constituent of milk of magnesia?
    [A] magnesium carbonate
    [B] magnesium hydroxide
    [C] light magnesium oxide
    [D] heavy magnesium oxide

19. Which of the following is the component of universal antidote?
    [A] magnesium dioxide
    [B] talc
    [C] kaolin
    [D] tannic acid

20. Standard solution used for limit tests of chlorides is ...
    [A]0.058% W/V NaCl
    [B] 0.58% W/V NaCl
    [C]0.09% W/V NaCl
    [D]1.0% W/V NaCl

SECTION – II

Q 2	Attempt any **FOUR** of the following	
A	Add a detailed note on copper used as essential and trace ion.	**05**
B	Draw and explain Jablonski Diagram.	**05**
C	Give classification of antidots and explain preparation and uses of sodium thiosulphate.	**05**
D	Give difference between thermal reaction and photochemical reactions.	**05**
E	Explain Joul Thompson effect.	**05**
F	Define and classify emetics and expectorants.	**05**

SECTION – III

Q 3	Attempt any **FOUR** of the following	
A	Define: radioactivity. Enlist methods for measurement of radioactivity. Discuss any one in detail.	**05**
B	Define and classify topical agents with suitable examples. Give preparation, assay and uses of zinc oxide.	**05**
C	Write role of fluoride in dental products and give assay and uses of sodium fluoride.	**05**
D	Explain beer-lambert law for photochemical reaction.	**05**
E	Define surface tension, viscosity. Discuss any one method for measurement of surface tension.	**05**
F	Explain the term limit test.give principle, chemical reaction involved in limit test for chloride.	**05**

Q 4	Attempt any **FOUR** of the following	
A	Add note on "factors affecting purity of pharmaceuticals."	**05**
B	Classify antimicrobial agents with examples and discuss their mechanism of action.	**05**
C	Discuss physiological role of oxygen and describe its method of preparation, properties and uses.	**05**
D	What is an adsorption isotherm? Discuss in detail Langmuir adsorption isotherm.	**05**
E	Define thermodynamics. Explain first law of thermodynamics.	**05**
F	Enumerate official compounds of iron. Give preparation and uses of ferrous sulphate.	**05**
